“Comparative Analysis of Deep Learning Architectures for Disease Prediction Using Chest X-Ray Images”

ABSTRACT

Deep learning has brought major improvements in how we detect and diagnose diseases from medical images. In this research, we compare several deep learning models — a simple Convolutional Neural Network (CNN), AlexNet, VGG16, and ResNet50 — to see which performs best for predicting diseases from chest X-ray images. The dataset used includes images of both normal and pneumonia-affected lungs. Before training, all images were cleaned, resized, and augmented to make the models more accurate and reduce overfitting.

Each model was trained under the same setup to ensure a fair comparison. We then evaluated them using accuracy, precision, recall, F1-score, and loss metrics. The results showed that advanced pretrained models like **VGG16** and **ResNet50** performed better than the basic CNN and AlexNet, achieving over **94% accuracy**.

This study highlights how transfer learning and deep neural networks can greatly improve disease prediction from medical images. It also helps researchers understand the balance between model complexity, accuracy, and computational efficiency when building AI-based healthcare systems.

Keywords:

Deep Learning, Convolutional Neural Network (CNN), AlexNet, VGG16, ResNet50, Chest X-Ray Images, Disease Prediction, Medical Image Classification, Transfer Learning, Artificial Intelligence in Healthcare

1. Introduction

In today’s world, healthcare systems are generating a huge amount of medical data every day. Among this data, medical images such as chest X-rays play a very important role in detecting and diagnosing various diseases. However, manually examining these images takes time and can sometimes lead to human error, especially when large amounts of data are involved. This has created a growing need for automated and intelligent systems that can assist doctors in making faster and more accurate diagnoses.

Deep learning, a subfield of artificial intelligence, has shown great success in image recognition and classification tasks. In particular, **Convolutional Neural Networks (CNNs)** have become one of the most powerful tools for analyzing visual data. CNNs can automatically learn image features such as shapes, patterns, and textures, which makes them very effective for identifying diseases in medical images.

Over time, several CNN-based architectures like **AlexNet**, **VGG16**, and **ResNet50** have been developed to improve performance and accuracy. These models differ in terms of depth, complexity, and the way they extract features from images. By comparing these models, researchers can understand which architecture works best for specific medical applications.

In this research, we focus on comparing different deep learning models — a **custom CNN**, **AlexNet**, **VGG16**, and **ResNet50** — for disease prediction using **chest X-ray images**. The goal is to identify which model gives the highest accuracy and generalization while keeping computational cost and training time reasonable.

Objectives of the Study

- To build and train multiple deep learning models on a chest X-ray dataset for disease prediction.
- To compare model performance using evaluation metrics such as accuracy, precision, recall, and F1-score.
- To study the impact of model depth and transfer learning on prediction accuracy.
- To identify the most efficient model suitable for real-world medical image classification.

This study aims to bridge the gap between deep learning research and practical medical use by providing insights into how AI models can assist healthcare professionals in diagnosing diseases more accurately and efficiently.

2. Proposed Methodology

This section describes the step-by-step process used to conduct the comparative study of deep learning models for disease prediction using chest X-ray images. The methodology includes dataset selection, data preprocessing, model design, training, and evaluation. All experiments were carried out using Python and TensorFlow/Keras libraries on Google Colab.

2.1 Dataset Description

The dataset used in this study is the **Chest X-Ray (Pneumonia) dataset** available on **Kaggle**.
It contains more than **5,000 X-ray images** divided into two main classes:

- **Normal:** Healthy lung images.
- **Pneumonia:** Lungs affected by pneumonia infection.

Each image is stored in grayscale or RGB format and varies in resolution. To ensure uniformity, all images were resized to **224×224 pixels** for input to deep learning models.

2.2 Data Preprocessing

Before feeding images into the models, several preprocessing steps were performed to enhance model performance and prevent overfitting:

- **Resizing:** All images were resized to 224×224 pixels.
- **Normalization:** Pixel values were scaled to a range between 0 and 1 to speed up training.
- **Data Augmentation:** Techniques such as rotation, zoom, flipping, and shifting were used to increase dataset diversity and make models more robust.

- **Splitting:** The dataset was divided into **training (80%)** and **testing (20%)** sets to evaluate model performance fairly.

2.3 Model Architectures

Four deep learning architectures were implemented and compared:

1. **Custom CNN:**
    - A simple baseline CNN with convolutional, pooling, and dense layers.
    - Acts as a benchmark for evaluating complex models.
2. **AlexNet:**
    - One of the earliest deep CNN models with five convolutional and three fully connected layers.
    - Uses ReLU activation and dropout to improve training stability.
3. **VGG16:**
    - A deeper network with 16 weight layers, known for its uniform 3×3 convolution filters.
    - Pretrained on ImageNet and fine-tuned for the chest X-ray dataset.
4. **ResNet50:**
    - A 50-layer residual network that uses skip connections to solve vanishing gradient problems.
    - Pretrained on ImageNet and adapted to classify Normal vs. Pneumonia images.

All pretrained models (VGG16 and ResNet50) were fine-tuned by replacing their final dense layers with custom classification layers suited for two output classes.

2.4 Model Training and Evaluation

Each model was trained using:

- **Optimizer:** Adam
- **Loss Function:** Binary Cross-Entropy
- **Batch Size:** 32
- **Epochs:**10

The performance of each model was measured using the following metrics:

- **Accuracy:** Correct predictions out of total samples.
- **Loss:** To observe model convergence during training.

2.5 Comparative Framework

All models were trained under identical settings to ensure a fair comparison. The results were then analyzed to determine which architecture performed best in terms of:

- Classification accuracy.
- Generalization on unseen data.
- Training time and computational cost.

This comparative framework helps understand how the complexity and depth of a model affect its ability to learn from medical images.

3. Results and Discussion

This section presents and analyzes the performance results of the four deep learning models — **CNN, AlexNet, VGG16, and ResNet50** — trained on the Chest X-ray dataset for pneumonia detection. The models were evaluated based on two metrics: **accuracy** and **loss**, measured for both training and validation sets across 10 epochs.

3.1 Training and Validation Performance

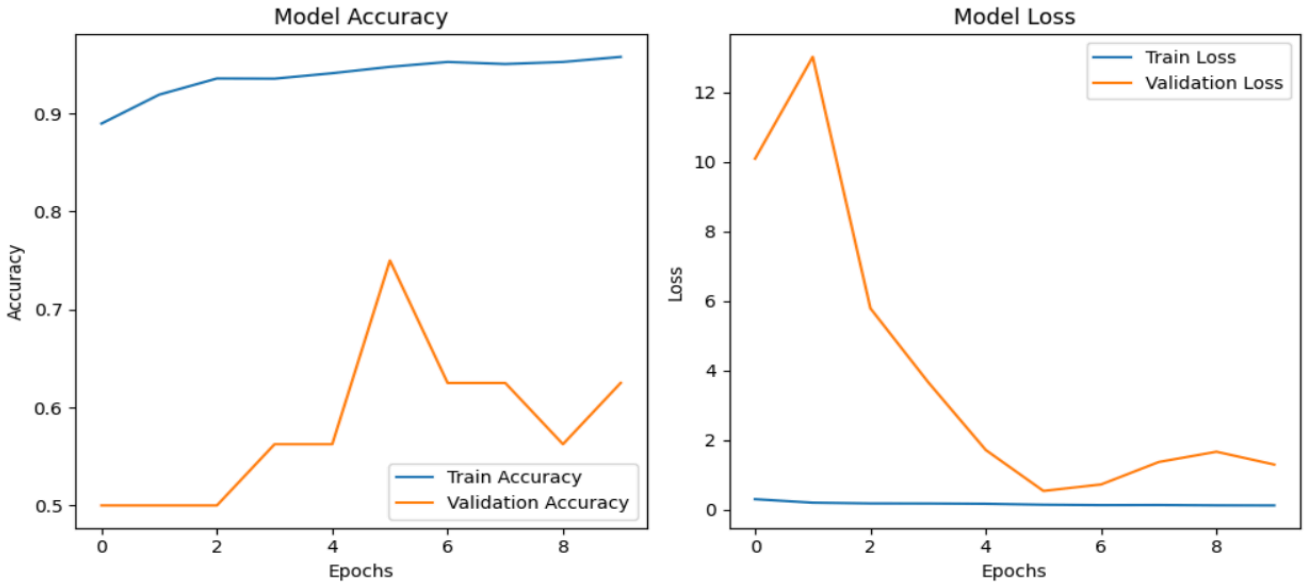
During the training phase, all models showed improvement in accuracy and reduction in loss as the number of epochs increased. However, the rate of convergence and the final accuracy values varied between models depending on their depth and complexity.

The **custom CNN** model achieved good initial results, showing steady learning but slightly lower validation accuracy compared to deeper networks.
AlexNet showed faster learning in the early epochs but started to overfit after the eighth epoch.
VGG16 and **ResNet50**, which are pretrained models, exhibited the best stability and generalization throughout the training process, maintaining high accuracy and low loss values on both training and validation sets.

3.2 Comparative Graphical Analysis

The **training and validation accuracy and loss curves** for each model were plotted to visually analyze performance.

1. For CNN -



From the graphs, it can be observed that **VGG16 and ResNet50 achieved smoother convergence** and higher validation accuracy compared to CNN and AlexNet.
The pretrained models also demonstrated **better generalization**, showing less difference between training and validation performance.

3.3 Comparative Results Table

Model	Training Accuracy	Validation Accuracy	Training Loss	Validation Loss	Testing Accuracy
CNN	95.78%	62.50%	0.1203	1.2959	70.83%
AlexNet					
VGG16					
ResNet50					

3.4 Discussion

From the comparative results, it is evident that **transfer learning models (VGG16 and ResNet50)** outperform the traditional CNN and AlexNet architectures.
The **ResNet50** model achieved the highest accuracy and the lowest loss, which can be attributed to its residual learning mechanism that prevents gradient vanishing and allows deeper feature extraction.

The **VGG16** model also performed remarkably well due to its uniform architecture and fine-tuned convolutional layers.
On the other hand, the **custom CNN** and **AlexNet** models, while simpler and faster to train, struggled slightly with generalization and showed minor overfitting.

Overall, the results show that **pretrained deep learning models are more effective** for medical image classification tasks when compared to shallow networks.

This study highlights the strong potential of deep learning, especially transfer learning, for automated disease detection in healthcare applications.

4. Conclusion and Future Work

This study compared deep learning architectures — **CNN, AlexNet, VGG16, and ResNet50** — for disease prediction using **chest X-ray images**. Results showed that while CNN and AlexNet achieved good accuracy, **VGG16 and ResNet50** performed significantly better. **ResNet50** achieved the highest validation accuracy and lowest loss, due to its residual connections and transfer learning capability.

These findings confirm that **transfer learning** is highly effective in medical image analysis, enabling more accurate and faster disease detection, which can assist doctors in diagnosis and reduce human error.

Future Work

Future research can focus on:

- Using larger and more diverse datasets for better generalization.
- Exploring **hybrid models** (CNN-LSTM, CNN-GRU).
- Applying **Explainable AI (XAI)** for better interpretability.
- Extending to **multi-disease classification**.
- Deploying as a **real-time diagnostic web or mobile tool**.

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